



INTE130 – Object-Oriented Programming Assignment.

(CyberShield system Project)

Project Report

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CyberShield System Project Report:

In Part A , the concept of a one person company and the role of digital agents in helping the founder manage the company are explained. It is also demonstrated how a single founder can manage a business using multiple systems and digital agents.

In part B the idea of the CyberShield project was presented as a cybersecurity company dedicated to **SMEs. Several digital agents have been proposed, such as :**

Threat monitoring agent.

Vulnerability agent.

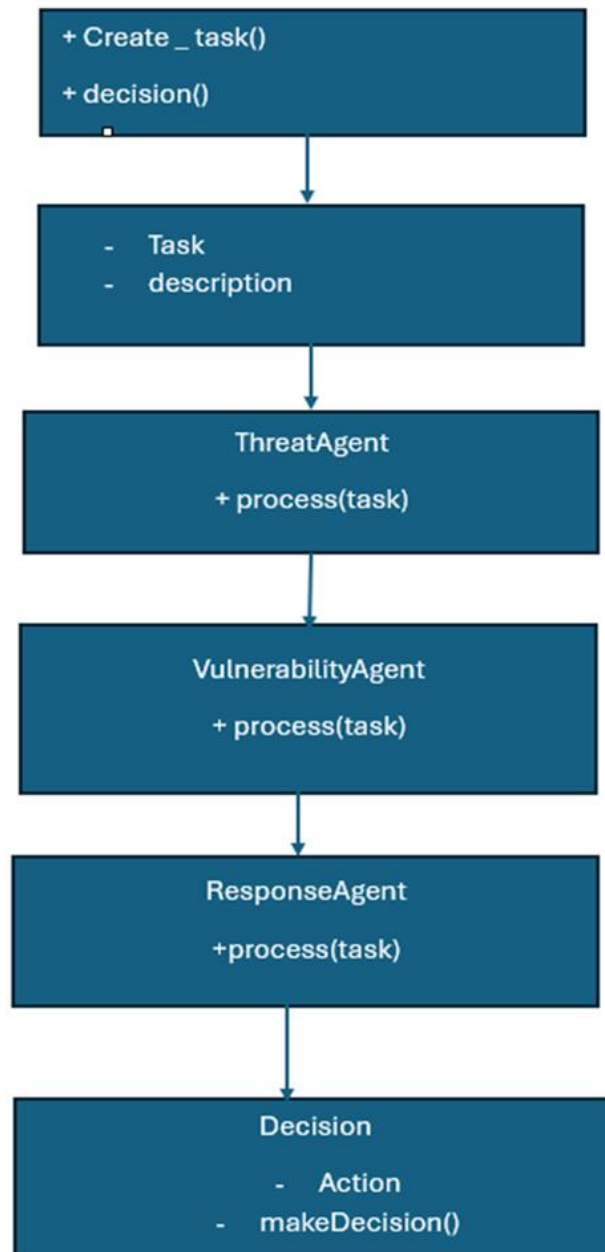
Response agent.

In part C , the CyberShield system was designed based on the same idea as in Part B , The System structure is agent and classes , and UML drawing were explained to clarify how the system works and the relationship between the founder and the digital agents.

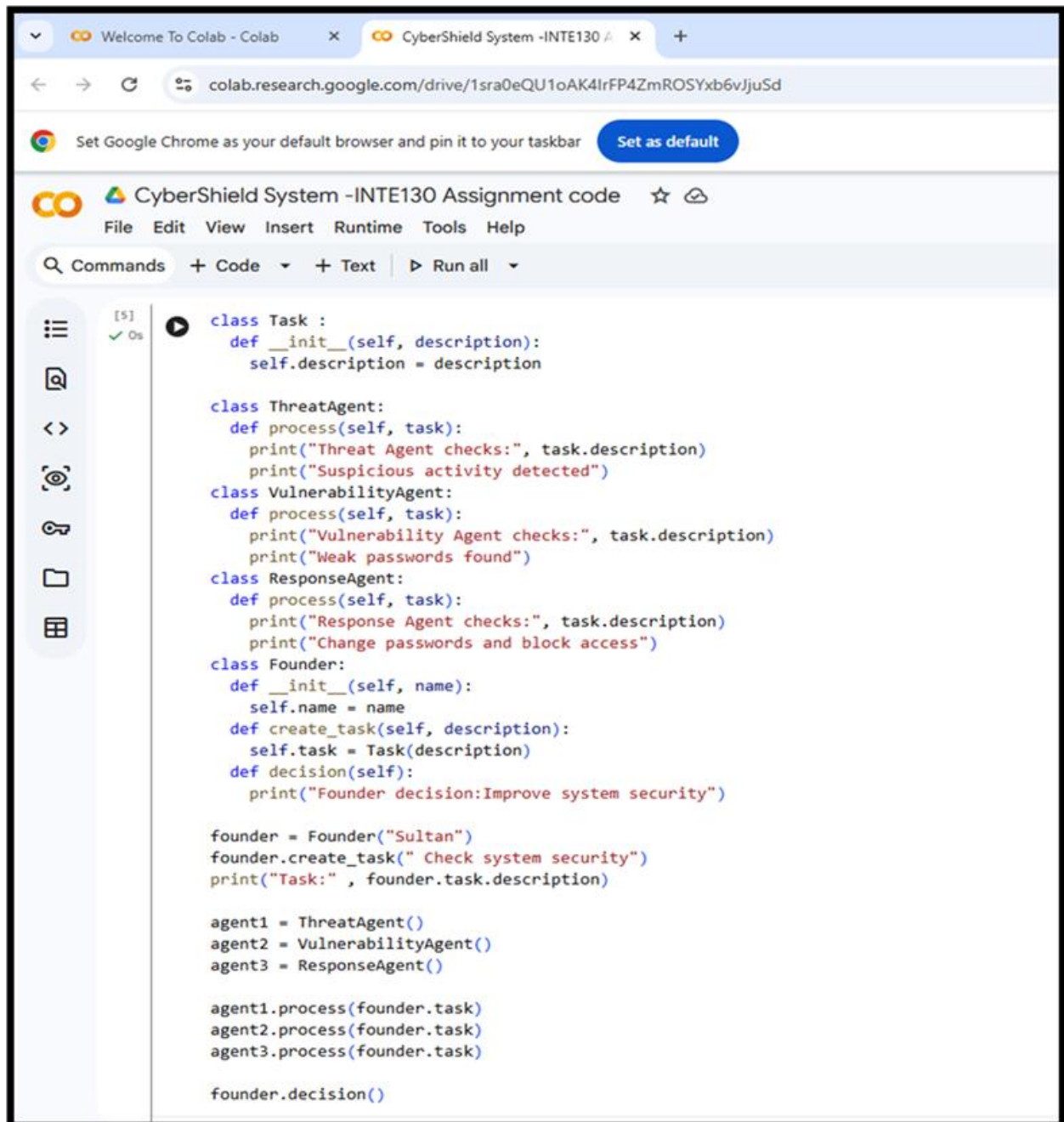
Finally , in part D the CyberShield system is implemented using python and oop concept. The same agents and work flow as in part c were used where the founder creates the tasks , the agents process the tasks , and then the founder makes the final decision.

This interconnection between parts A , B , C and D makes the project organized and clear and appears as an integrated and interconnected cybersecurity system.

-UML Daigram:



System code:



```
[5] ✓ Os
class Task :
    def __init__(self, description):
        self.description = description

class ThreatAgent:
    def process(self, task):
        print("Threat Agent checks:", task.description)
        print("Suspicious activity detected")
class VulnerabilityAgent:
    def process(self, task):
        print("Vulnerability Agent checks:", task.description)
        print("Weak passwords found")
class ResponseAgent:
    def process(self, task):
        print("Response Agent checks:", task.description)
        print("Change passwords and block access")
class Founder:
    def __init__(self, name):
        self.name = name
    def create_task(self, description):
        self.task = Task(description)
    def decision(self):
        print("Founder decision:Improve system security")

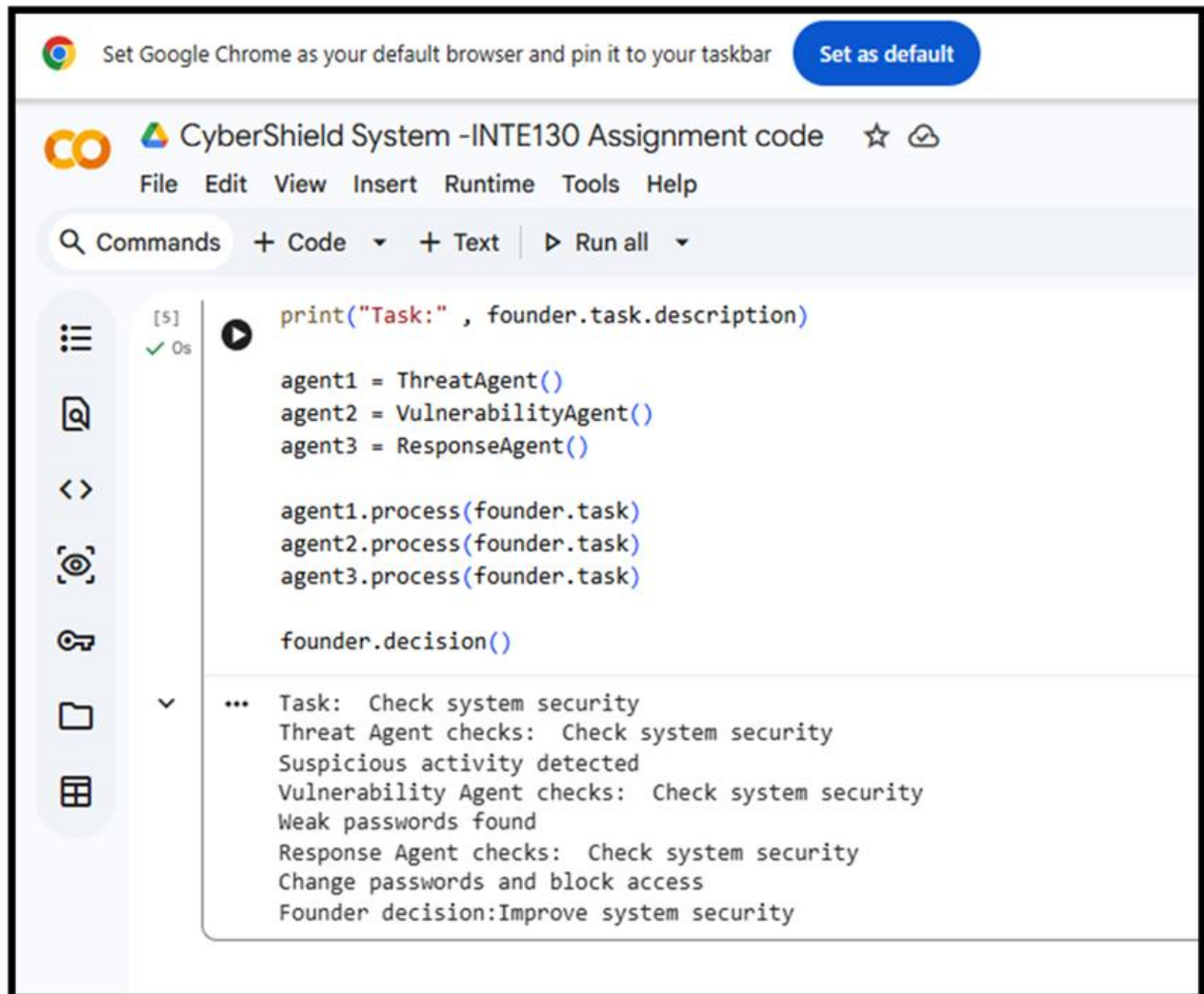
founder = Founder("Sultan")
founder.create_task(" Check system security")
print("Task:" , founder.task.description)

agent1 = ThreatAgent()
agent2 = VulnerabilityAgent()
agent3 = ResponseAgent()

agent1.process(founder.task)
agent2.process(founder.task)
agent3.process(founder.task)

founder.decision()
```

Output Code:



The screenshot shows a web browser window with a notification at the top: "Set Google Chrome as your default browser and pin it to your taskbar" with a "Set as default" button. The browser address bar shows "CyberShield System -INTE130 Assignment code" with star and cloud icons. Below the address bar is a menu bar with "File", "Edit", "View", "Insert", "Runtime", "Tools", and "Help". A toolbar contains a search icon, "Commands", "+ Code", "+ Text", "Run all", and a dropdown arrow. On the left is a sidebar with icons for a menu, search, code editor, runtime, key, folder, and a grid. The main area is split into two panes. The top pane shows Python code with a play button icon and a status "[5] ✓ 0s". The code defines three agents (ThreatAgent, VulnerabilityAgent, ResponseAgent), processes a task for each, and then makes a decision. The bottom pane shows the output of the code execution, which is a series of text lines describing the task and the actions taken by the agents.

```
print("Task:" , founder.task.description)

agent1 = ThreatAgent()
agent2 = VulnerabilityAgent()
agent3 = ResponseAgent()

agent1.process(founder.task)
agent2.process(founder.task)
agent3.process(founder.task)

founder.decision()
```

```
... Task: Check system security
Threat Agent checks: Check system security
Suspicious activity detected
Vulnerability Agent checks: Check system security
Weak passwords found
Response Agent checks: Check system security
Change passwords and block access
Founder decision:Improve system security
```